

Akhil Chigurupati

Austin, TX | (720) 655-2444 | chigurupati.akhil5@gmail.com | [LinkedIn](#)

PROFESSIONAL SUMMARY

Software engineer specializing in distributed backend systems — from designing Python microservices and REST APIs to building real-time Kafka streaming pipelines and deploying containerized services with Docker and Kubernetes. Experienced in full-stack development with a focus on system observability, operational efficiency, and scalable architecture across cloud-native environments.

EDUCATION

Regis University

Master's, Information Systems — Denver, CO, USA

PROFESSIONAL EXPERIENCE

Tata Consultancy Services | Apr 2023 – May 2024

Software Engineer

- Built Python microservices and REST APIs for an AI-driven infrastructure monitoring platform managing 500+ production services, improving modularity and reducing debugging overhead across distributed systems.
- Designed and deployed containerized applications using Docker and Kubernetes, managing 1,000+ pods with auto-scaling, load balancing, and high availability across distributed clusters.
- Architected Kafka-based real-time streaming pipelines processing 50,000+ events/min for log and metrics ingestion, enabling near real-time observability and faster incident detection.
- Developed AI/ML anomaly detection models integrated directly into monitoring workflows, enabling proactive identification of system failures and performance degradation before user impact.
- Automated end-to-end remediation workflows and led design and code reviews, cutting manual incident resolution effort by 30–40% and significantly improving mean time to recovery.

PROJECT EXPERIENCE

URL Shortener — Distributed Analytics Platform

- Built a full-stack URL shortener with a real-time analytics pipeline using FastAPI, PostgreSQL, Redis, and Kafka — decoupling click tracking from the redirect path via an async Kafka consumer worker.
- Implemented cache-aside pattern with Redis for sub-millisecond slug lookups and instrumented the API with Prometheus metrics visualized on a live Grafana dashboard.
- Containerized the full stack with Docker Compose and automated code quality checks via a GitHub Actions CI pipeline running pytest and linting on every push.

Review Reality — Fake Review Detection System (Published)

- Built and optimized a fake review detection system in Python achieving 92% precision across 10 major review platforms, leveraging ensemble ML algorithms and advanced feature engineering on 500,000+ review entries.
- Reduced false positives by 30% through targeted feature extraction and preprocessing pipelines, improving model training efficiency and increasing fraudulent review identification by 15%.
- Published research findings, validating the model's effectiveness in improving review trustworthiness at scale.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, Java, HTML, CSS, SQL

Backend Technologies: FastAPI, REST APIs, Microservices

Frontend Technologies: React.js, Vite

AI/ML Technologies: Scikit-learn, TensorFlow, Anomaly Detection, Predictive Analytics

Containerization & Orchestration: Docker, Kubernetes, Helm

Messaging & Streaming: Apache Kafka

Monitoring & Logging: Prometheus, Grafana, ELK Stack

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: AWS, GitHub Actions, GitLab CI/CD, Linux

Technical Skills: Data Structures, System Design, Distributed Systems, API Development, Root Cause Analysis, Infrastructure Monitoring, CI/CD Pipelines, Debugging, Algorithms